Implementation of a Luenberger Observer and a Linear Kalman Filter to Estimate the Current and Speed of a DC Motor

Andrés Ramírez Simón Patiño Jeronimo Valencia
Jorge Alejandro de Santiago
Mechatronics Engineering – Digital Control
Universidad EIA, Envigado, Colombia

***Abstract*—Two discrete state observers, a full-order predictor Luenberger observer and a linear Kalman filter, are designed and implemented to estimate the armature current and the speed of a Pololu 37D #4753 (50:1) gearmotor, measuring only the speed with an incremental encoder. The parametric model from the previous laboratory is tuned so that it draws $0.2\,\mathrm{A}$ and reaches the nominal speed at $12\,\mathrm{V}$, and it is refined with steps measured on the elbow joint of a SCARA robot. Both observers run on a Raspberry Pi Pico with a $10\,\mathrm{ms}$ sampling period and are evaluated under steps, reference changes and load disturbances. Both give consistent current estimates; under disturbances the Kalman filter tracks the speed with a $7.8\,\mathrm{rad/s}$ RMS error, against $34.8\,\mathrm{rad/s}$ for the Luenberger observer.**

***Index Terms*—Luenberger observer, Kalman filter, DC motor, state space, Raspberry Pi Pico.**

I. Introduction

In many applications not every variable of a system can be measured, either because no sensor is available or because measuring it is expensive. In a DC motor the speed is easily obtained with an encoder, but the armature current requires an additional sensor. A state observer reconstructs the unmeasured states from the system model, the input and the measured outputs [4].

This work implements the two observers proposed in the laboratory guide [1], [2]: the Luenberger observer, designed by pole placement with Ackermann's formula, and the linear Kalman filter, designed from the model and measurement uncertainty [5]. Unlike the guide, which uses a myRIO board and LabVIEW, the implementation runs on a Raspberry Pi Pico driving the elbow motor of a SCARA robot.

II. Theoretical Framework

A. DC Motor Model

With the states $x = [i_a \;\; \omega]^T$, the input $u = V_a$ and the measured output $y = \omega$, the motor equations are

$$\dot{i}_a = -\frac{R_a}{L_a} i_a - \frac{K_v}{L_a}\omega + \frac{1}{L_a} V_a, \tag{1}$$

$$\dot{\omega} = \frac{K_v}{J} i_a - \frac{\beta}{J}\omega, \tag{2}$$

that is, $\dot{x} = A_c x + B_c u$, $y = Cx$ with

$$A_c = \begin{bmatrix} -\frac{R_a}{L_a} & -\frac{K_v}{L_a} \\ \frac{K_v}{J} & -\frac{\beta}{J} \end{bmatrix}, \quad B_c = \begin{bmatrix} \frac{1}{L_a} \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix}. \tag{3}$$

B. Discretization

With a zero-order hold and sampling period T_s:

$$A = e^{A_c T_s}, \qquad B = \int_0^{T_s} e^{A_c \tau}\, d\tau\, B_c, \tag{4}$$

$$x_k = A x_{k-1} + B u_{k-1}, \qquad y_k = C x_k. \tag{5}$$

C. Luenberger Observer

The full-order predictor observer uses the input and the measurement from the previous sample:

$$\hat{x}_k = (A - LC)\hat{x}_{k-1} + L\, y_{k-1} + B\, u_{k-1}. \tag{6}$$

The estimation error evolves as $e_k = (A - LC)e_{k-1}$, so L is chosen to place the eigenvalues of $A - LC$ inside the unit circle. With Ackermann's formula,

$$L = \Phi(A)\, \mathcal{O}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \mathcal{O} = \begin{bmatrix} C \\ CA \end{bmatrix}, \tag{7}$$

where $\Phi(z)$ is the desired characteristic polynomial.

D. Linear Kalman Filter

Starting from $\hat{x}_0$ and P_0, each sample performs a prediction and a correction [5]:

$$\hat{x}_k^- = A\hat{x}_{k-1}^+ + B u_{k-1}, \qquad P_k^- = A P_{k-1}^+ A^T + Q, \tag{8}$$

$$K_k = P_k^- C^T (C P_k^- C^T + R)^{-1}, \tag{9}$$

$$\hat{x}_k^+ = \hat{x}_k^- + K_k(y_k - C\hat{x}_k^-), \qquad P_k^+ = (I - K_k C) P_k^-, \tag{10}$$

where Q represents the model uncertainty and R the measurement variance.

Table I
MOTOR PARAMETERS

Parameter	Lab 1	Tuned
R_a (Ω)	72.7	2.18
L_a (H)	6.62×10^{-2}	5.56×10^{-4}
K_v (V s/rad)	1.06×10^{-2}	1.104×10^{-2}
J (kg m^2)	8.55×10^{-7}	2.81×10^{-6}
β (N m s)	1.59×10^{-6}	2.11×10^{-6}
i_a at 12 V (A)	0.084	0.200
ω at 12 V (rad/s)	558	1047
τ_{mech} (ms)	273	48

III. EXPERIMENTAL SETUP

The elbow motor of a SCARA RRP robot is used: a 12 V Pololu 37D #4753 gearmotor with a 50:1 gearbox [3], which drives the forearm through a 3:1 pulley. The Hall encoder gives 64 counts per revolution of the motor shaft. The speed is computed as the position difference between two samples divided by T_s, so its resolution is $2\pi/(64\,T_s) \approx 9.8\,\mathrm{rad/s}$. The motor is driven by a BTS7960 H-bridge with a 20 kHz PWM from a Raspberry Pi Pico, which reads the encoder with a PIO state machine.

The firmware runs a loop every $T_s = 10\,\mathrm{ms}$ that measures the speed, updates both observers and applies the experiment voltage; the data are sent over USB and a Python script stores and plots them. Since the elbow has a limited travel (about 3.7 rad), the firmware homes against the limit switch, cuts the voltage before either end of travel and stops the motor if it stalls. The shoulder and the prismatic axis are kept braked.

IV. MOTOR MODEL

A. Laboratory 1 Model

The previous parametric identification gave the parameters in the Lab 1 column of Table I, with an armature resistance of $72.7\,\Omega$. With that model the motor reaches 0.084 A and $558\,\mathrm{rad/s}$ (107 rpm at the output) at 12 V, far from the 0.2 A and 200 rpm specified by the manufacturer. K_v and β were consistent with the datasheet, but R_a was about 30 times larger than the value derived from the stall current (5.5 A at 12 V).

B. Model Tuning

R_a was set to $12/5.5 = 2.18\,\Omega$, and the two steady-state conditions of the guide were imposed, $i_a = 0.2\,\mathrm{A}$ and $\omega = \omega_{nom} = 1047\,\mathrm{rad/s}$ (10 000 rpm at the motor shaft), which gives

$$K_v = \frac{12 - R_a\, i_a}{\omega_{nom}}, \qquad \beta = \frac{K_v\, i_a}{\omega_{nom}}. \tag{11}$$

J and L_a were then fitted by least squares to reproduce three steps of 4, 6 and 8 V measured on the elbow. The resulting J includes the forearm inertia reflected to the motor shaft. Table I summarizes the parameters.

With these values,

$$A_c = \begin{bmatrix} -3924.1 & -19.86 \\ 3935.3 & -0.752 \end{bmatrix}, \qquad B_c = \begin{bmatrix} 1798.6 \\ 0 \end{bmatrix}, \tag{12}$$

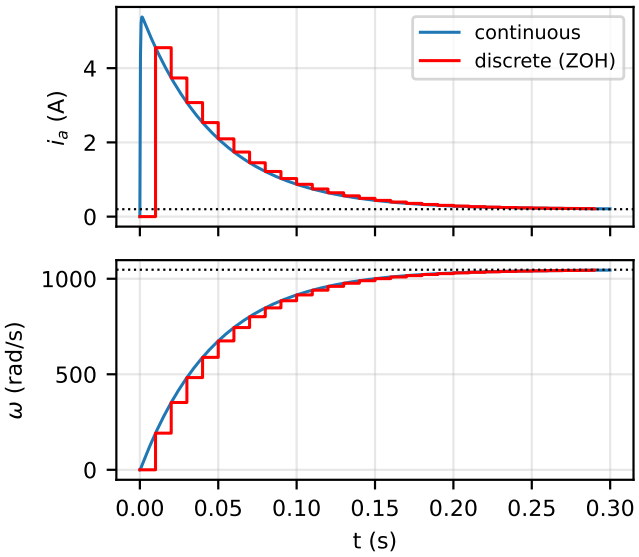


Figure 1. Response of the tuned model to 12 V: the current settles at 0.2 A and the speed at $1047\,\mathrm{rad/s}$.

and with $T_s = 10\,\mathrm{ms}$,

$$A = \begin{bmatrix} -0.00419 & -0.00415 \\ 0.8233 & 0.8166 \end{bmatrix}, \qquad B = \begin{bmatrix} 0.3793 \\ 15.991 \end{bmatrix}. \tag{13}$$

The discrete plant poles are $z \approx 0$ (electrical mode, $\tau_e = L_a/R_a \approx 0.25\,\mathrm{ms} \ll T_s$) and $z = 0.812$ (mechanical mode). Fig. 1 verifies the nominal condition and Fig. 2 compares the model with the data.

The tuned model reproduces the measured time constant, but overestimates the speed by about $40\,\mathrm{rad/s}$ at 4 V. The measured steady-state speeds (305, 496 and $687\,\mathrm{rad/s}$) grow by about $96\,\mathrm{rad/s}$ per volt, and extrapolated to 12 V they give $\approx 1070\,\mathrm{rad/s}$, close to the nominal speed. However, the line does not pass through the origin: there is a dead zone of about 0.8 V caused by the dry friction of the gearbox and the arm, which a linear model cannot represent.

V. OBSERVER DESIGN

A. Luenberger

The observability matrix $\mathcal{O} = [C;\, CA]$ has rank 2, so the system is observable from the speed measurement alone. The initial design, with poles at $z = 0.3$ and 0.4, produced negative current estimates down to -3.5 A with the real data (Fig. 3). The reason is that the electrical pole of the plant is already at $z \approx 0$: moving it to 0.3 makes the current mode of the observer slower than that of the plant, and the small model error (the friction) accumulates in the current. The poles $z = \{0,\ 0.2\}$ were therefore chosen, which keep the electrical pole and speed up the mechanical one, giving

$$\Phi(z) = z^2 - 0.2z, \qquad L = \begin{bmatrix} -0.00312 \\ 0.6124 \end{bmatrix}. \tag{14}$$

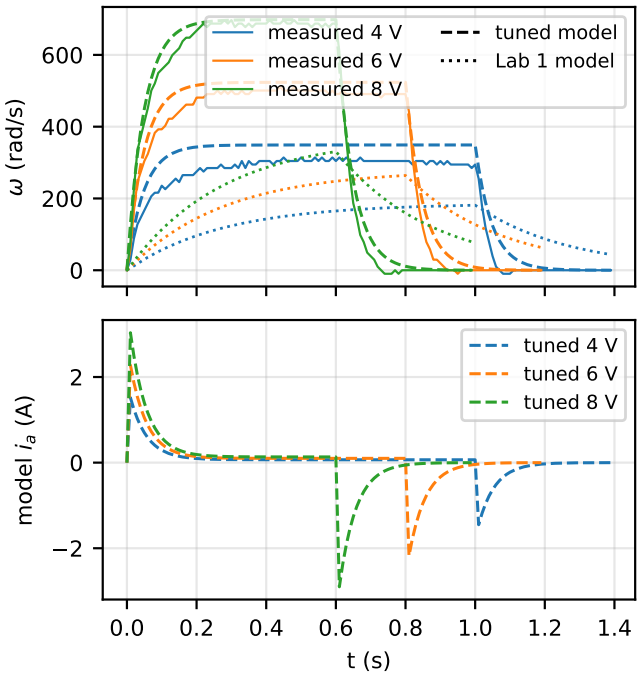


Figure 2. Steps of 4, 6 and 8 V measured on the elbow, compared with the Lab 1 model (dotted) and with the tuned model (dashed).

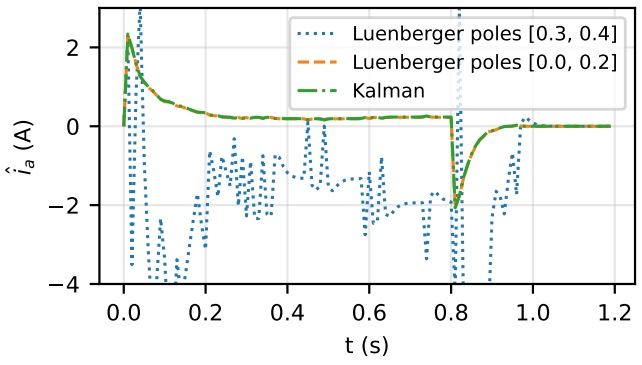


Figure 3. Estimated current with the data of the 6 V step: with the poles $\{0.3, 0.4\}$ the Luenberger observer gives negative currents; with $\{0, 0.2\}$ it matches the Kalman filter.

B. Kalman

The measurement variance was taken from the steady-state data of the steps, between 26 and 45 $(\mathrm{rad/s})^2$, and $R = 35$ was used. For the model, an uncertainty of 10 mA in the current and 10 rad/s in the speed was assumed, and the initial estimate was given a larger uncertainty in the current, which is not measured:

$$Q = \begin{bmatrix} 10^{-4} & 0 \\ 0 & 100 \end{bmatrix}, \quad P_0 = \begin{bmatrix} 0.1 & 0 \\ 0 & 100 \end{bmatrix}. \tag{15}$$

The gain converges to $K = [-5.98 \times 10^{-4} \quad 0.771]^T$. No retuning was needed with these values.

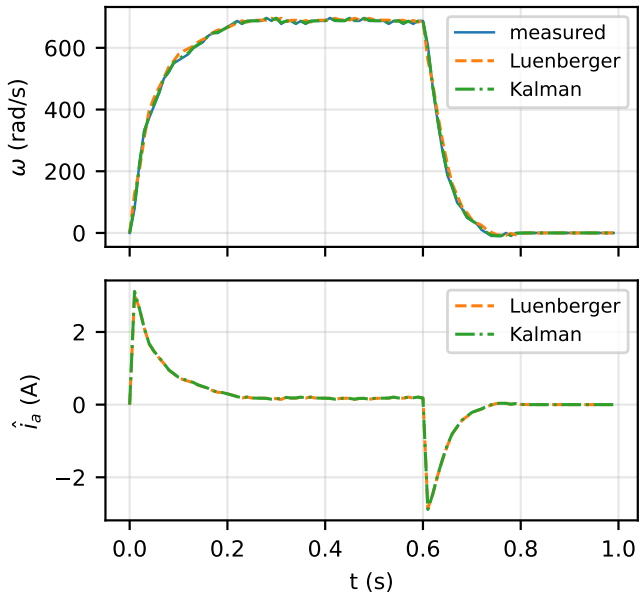


Figure 4. Simulation of the observers on the measured 8 V step.

VI. Implementation

The matrices are computed in Python and written to a header file that the firmware compiles. Since $C = [0\ 1]$, the innovation is a scalar and the Kalman filter needs no matrix inversion. The Luenberger observer step is:

```
// x(k) = (A - L C) x(k-1) + L y(k-1) + B u(k-1)
float e = y_ant - x[1];
float x0 = AD[0]*x[0] + AD[1]*x[1] + BD[0]*u_ant + LG[0]*e;
float x1 = AD[2]*x[0] + AD[3]*x[1] + BD[1]*u_ant + LG[1]*e;
```

and the Kalman filter correction:

```
float s = Pm[3] + RK; // C P- C' + R
float K0 = Pm[1]/s, K1 = Pm[3]/s; // gain
float e = y - x1; // innovation
x[0] = x0 + K0*e; x[1] = x1 + K1*e;
```

The 10 ms loop includes the measurement, both observers, the actuation and the data transmission.

VII. Results

A. Simulation with Measured Data

Before the implementation, both observers were run on the measured steps with the same algorithm as the firmware (Fig. 4). Over the three steps the speed RMS error was 12–14 rad/s for Luenberger and 2.6–3.1 rad/s for Kalman. The estimated steady-state currents agreed between both methods (0.27, 0.23 and 0.19 A at 4, 6 and 8 V).

B. Implementation on the Pico

Fig. 5 shows a 5 V step estimated in real time. Both observers follow the measured speed, and the estimated current shows the start-up peak (1.9 A), settles at ≈0.25 A and becomes negative during braking.

Under reference changes of 4, 8, 3, −6 and −3 V (Fig. 6), the observers follow the speed in every segment, including

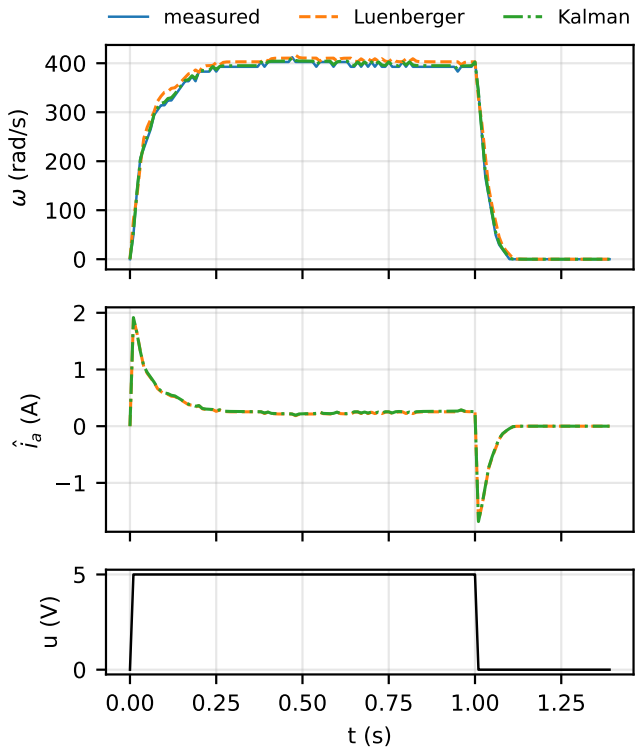


Figure 5. Implementation on the Pico: 5 V step for 1 s.

Table II
RMS Error of the Estimated Speed (rad/s)

Test	Luenberger	Kalman
4, 6 and 8 V steps (simulation)	12–14	2.6–3.1
Load disturbance (Pico)	34.8	7.8

the direction reversal. The estimated current shows a peak at each reference change and reaches -3 A when the direction reverses, which is the expected behavior of a DC motor.

For the disturbance test, the elbow moves back and forth at ± 3.5 V for 6 s, reversing at 0.5 and 1.4 rad, while the forearm is braked by hand (Fig. 7). Each time the load is applied, the measured speed drops and the estimated current rises from ≈ 0.4 A to 1.0–1.3 A. That value approaches the stall current at 3.5 V ($3.5/R_a \approx 1.6$ A). Table II summarizes the errors.

VIII. Discussion

Tuning the model was necessary: the Lab 1 model did not meet the nominal condition because its armature resistance was about 30 times larger than expected. By fixing K_v and β from the steady-state conditions and fitting only J and L_a, the model meets the manufacturer's specification and reproduces the measured dynamics. The remaining error is due to dry friction, which is not linear.

For the Luenberger observer, the choice of poles turned out to be decisive. With $T_s = 10$ ms the electrical mode of the plant is practically instantaneous, and an observer pole far

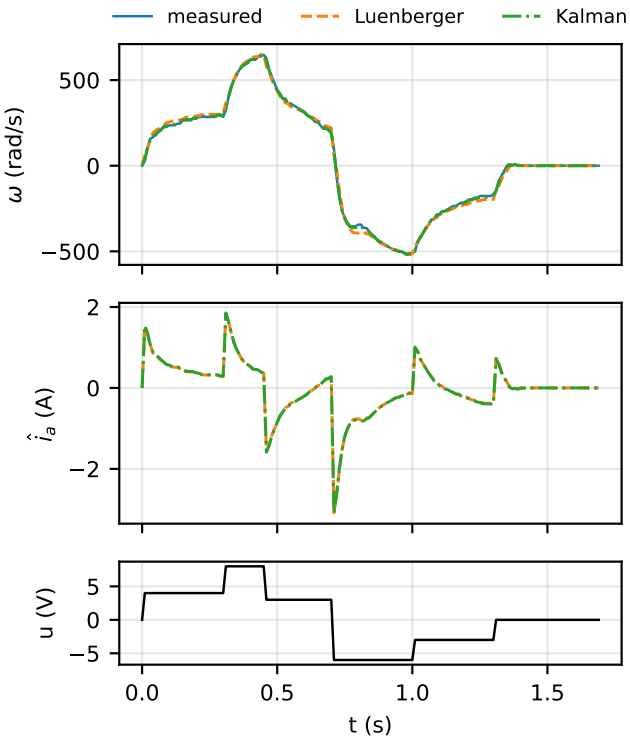


Figure 6. Reference changes: 4, 8, 3, -6, -3 and 0 V.

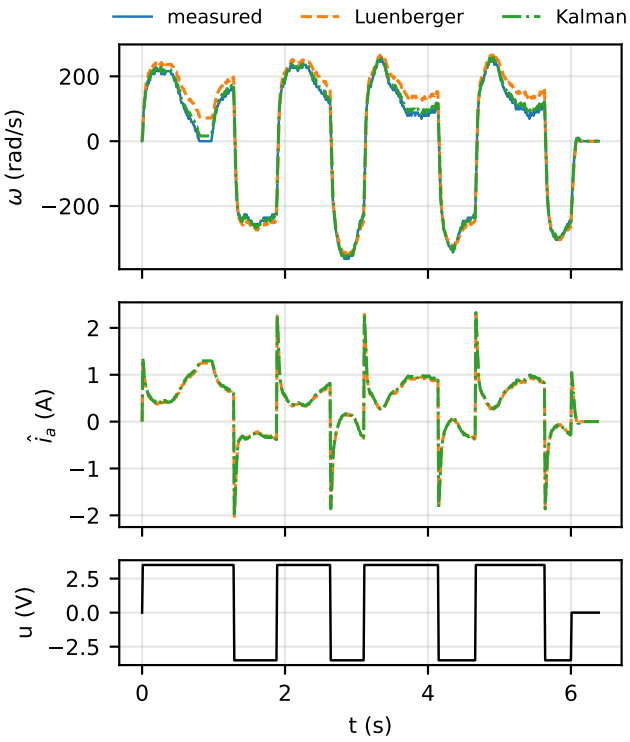


Figure 7. Disturbance: the elbow moves back and forth at ± 3.5 V and is braked by hand three times.

from $z = 0$ makes the current estimate very sensitive to model errors. The Kalman filter, on the other hand, computes its gain from Q and R without choosing poles by hand, and it behaved well from the first design.

With the motor braked, the estimated speed does not reach zero (Fig. 7): neither model includes the load torque, so the observer expects the motor to turn whenever voltage is applied. Kalman suffers less from this bias because its gain on the speed (0.77) gives more weight to the measurement. A natural improvement would be to add the load torque as a third state.

A limitation of the setup also appeared in practice: when the direction was reversed with the motor at high speed, the current peak and the energy returned by the motor shut down the power supply. The experiments were therefore limited to reversals of $\pm 6\,\mathrm{V}$ or less in the middle of the elbow travel.

IX. Conclusions

- With a single speed measurement, both observers reconstruct the armature current consistently, in steady state, in transients and under load.
- Under disturbances, the Kalman filter tracks the speed about four times better than the Luenberger observer (7.8 versus $34.8\,\mathrm{rad/s}$ RMS) and filters the encoder quantization noise.
- The Luenberger design depends on the chosen poles. Modes much faster than the sampling period should be kept at $z \approx 0$.
- The Raspberry Pi Pico runs both observers and the data acquisition in a $10\,\mathrm{ms}$ loop without difficulty, which makes it a low-cost alternative to the myRIO board.

References

[1] M. Giraldo, "Laboratory 4: State observers and estimators," Digital Control, Universidad EIA, 2026.

[2] J. Cossio, S. Neumann and M. A. Giraldo, "Guía práctica para la implementación de observadores de estado discretos," Universidad EIA.

[3] Pololu Corporation, "37D Metal Gearmotors," datasheet, rev. 1.2.

[4] K. Ogata, *Modern Control Engineering*, 5th ed. Prentice Hall, 2010.

[5] G. Welch and G. Bishop, "An introduction to the Kalman filter," UNC Chapel Hill, 1995.